

ROLLING MILL OPERATIONS MANUAL

Document ID: MAN-RM-001

Equipment Description

Rolling mills reduce steel thickness through rotating rolls and high mechanical forces.

Normal Operating Parameters

- Roll Temperature: 60°C - 100°C
- Vibration: 3 - 6 mm/s
- Motor Current: 100 - 250 A

Critical Thresholds

- Vibration > 8 mm/s
- Bearing Temperature > 85°C
- Current > 300 A

Common Failure Modes

1. Bearing Wear
2. Shaft Misalignment
3. Gearbox Failure
4. Lubrication Failure

Troubleshooting Guide

High Vibration

Possible Causes:

- Bearing degradation
- Misalignment
- Roll imbalance

Actions:

- Inspect bearings
- Perform alignment check

High Motor Current

Possible Causes:

- Mechanical overload

- Drive train resistance

Actions:

- Inspect gearbox
- Verify roll condition

Maintenance Recommendations

Daily:

- Monitor vibration

Weekly:

- Lubrication inspection

Monthly:

- Alignment verification

Failure Signature

Bearing Wear Pattern:

- Gradual vibration increase
- Rising bearing temperature
- Increased motor current

Recommended Action:

Replace bearing within 7 days if vibration exceeds 8 mm/s.